

| REQ ID  | TYPE           | DESCRIPTION                                                                                                                                                                             | PRIORITY/TYPE                                 | ACCEPTANCE CRITERIA                                                                                                                                                                                                                                                 | RATIONALE                                                                         |  |
|---------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--|
| FR-001  | functional     | The system shall validate proposed pallet placement coordinates against the rack maximum load capacity before confirming storage                                                        | high                                          | Pass: pallet assigned to valid rack;<br>Fail: Placement permitted on rack where total weight exceeds rated structural limit.                                                                                                                                        | Ensure safe storage by preventing rack overloading                                |  |
| FR-002  | functional     | The system shall allow a Warehouse Operator to scan a pallet barcode or RFID tag to identify the pallet and retrieve its stored inventory details.                                      | high                                          | Pass: A valid barcode/RFID scan retrieves the correct pallet details within 2 seconds. Fail: The system retrieves incorrect details or fails to identify a valid scanned pallet.                                                                                    | Ensures accurate identification and tracking of warehouse inventory.              |  |
| FR-003  | functional     | The system shall record and update the 3D storage coordinates of a pallet whenever it is moved to a different shelf location.                                                           | high                                          | Pass: After a pallet movement is confirmed, the system displays the new row, column, and level coordinates and logs the movement. Fail: The old or incorrect coordinates remain associated with the pallet.                                                         | Maintains accurate pallet location information for efficient inventory retrieval. |  |
| FR-004  | functional     | The system shall allow the Logistics Supervisor to view the current total weight stored on each rack and receive an alert when the weight reaches or exceeds the maximum load capacity. | medium                                        | Pass: The system displays the rack's current load and maximum capacity and generates an alert when the defined weight threshold is reached or exceeded. Fail: No alert is generated when the rack exceeds the defined limit.                                        | Helps supervisors monitor rack loading and prevent unsafe storage conditions.     |  |
| FR-005  | functional     | The system shall allow a Warehouse Operator to search for a pallet using its SKU or pallet ID and display its current storage coordinates.                                              | low                                           | Pass: When a valid SKU or pallet ID is entered, the system displays the corresponding pallet's current rack, row, column, and level coordinates. Fail: The system returns an incorrect location or does not return a location for an existing pallet.               | Improves the efficiency of locating specific inventory items.                     |  |
| NFR-001 | non-functional | The warehouse item lookup service must locate any SKU pallet coordinate across 100,000 bins in under 100 ms.                                                                            | performance and security, high priority       | Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Benchmarking tests show that the item lookup takes 100 ms or longer, or the system fails to meet the required security standards under simulated peak load. | Ensures fast, secure inventory lookup during peak warehouse operations.           |  |
| NFR-002 | non-functional | The system shall maintain an availability of at least 99.5% during warehouse operating hours, excluding scheduled maintenance.                                                          | medium priority, availability and reliability | Pass: System monitoring confirms at least 99.5% uptime during operating hours over a 30-day period. Fail: Availability falls below 99.5%.                                                                                                                           | Ensures warehouse operations can continue with minimal disruption.                |  |
|         |                |                                                                                                                                                                                         |                                               |                                                                                                                                                                                                                                                                     |                                                                                   |  |